INCH-POUND

MIL-C-70532 (AR)
AMENDMENT 8
18 December 1998
SUPERSEDING
AMENDMENT 7
30 July 1997

MILITARY SPECIFICATION

CARTRIDGES, 105MM, HE, RA,XM913 AND XM927 (WITH WARHEAD AND MOTOR BODY) LOADING, ASSEMBLING AND PACKING

MIL-C-70532 was inactivated after 21 August 1998 for new design.

This amendment forms a part of Military Specification MIL-C-70532 (AR), dated 29 October 1990, and is approved for use by the U.S. Army Armaments, Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Change "XM913" to "M913" (and wherever it appears).
Change "XM927" to "M927" (and wherever it appears).

PAGE 2

2.1.2: DRAWINGS

Change "(See 6.9)" to "(See 6.11)".
Change "9290990" to "9390990".

Dwg. 12624495 Add: "PA111," after "assembly".

Dwg. 12624496 Change: "Ammunition" to "Ammo".

Dwg. 12934903 Change: "XM912" to "M913".

Dwg. 12630699: Delete in its entirety

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Drawing 19-48-4231/45-20PM1006: Add "and" after "Ammunitions" and delete "Cyclindrical" and substitute "Cylindrical".

PAGE 3

3.1.1, Line 2, Change "operation" to "operations".

3.2.3, Change "grains" to "grain" two places.

PAGE 4

3.2.3 Delete in its entirety.

3.2.4 Delete in its entirety.

PAGE 5

3.3.5 Change: "thousands" to "thousandths".

PAGE 6

Delete “Figure 1.” And substitute the following included “Figure 1.”.

PAGE 7

3.6 Capitalize: "proving" in title.

PAGE 8

* Page 8, Delete 3.6.1.2.3 in its entirety and substitute:

“3.6.1.2.3 M913 cartridge muzzle velocity. The average muzzle velocity for cartridges fired at 70 degrees F shall be not less than 643 meters per second (mps) nor greater than 659 mps and the standard deviation shall not exceed 2.44 mps.”

3.6.2.1, Change title from “Pressure.” To “M913 cartridge pressure.”

* Add 3.6.2.2 as follows:

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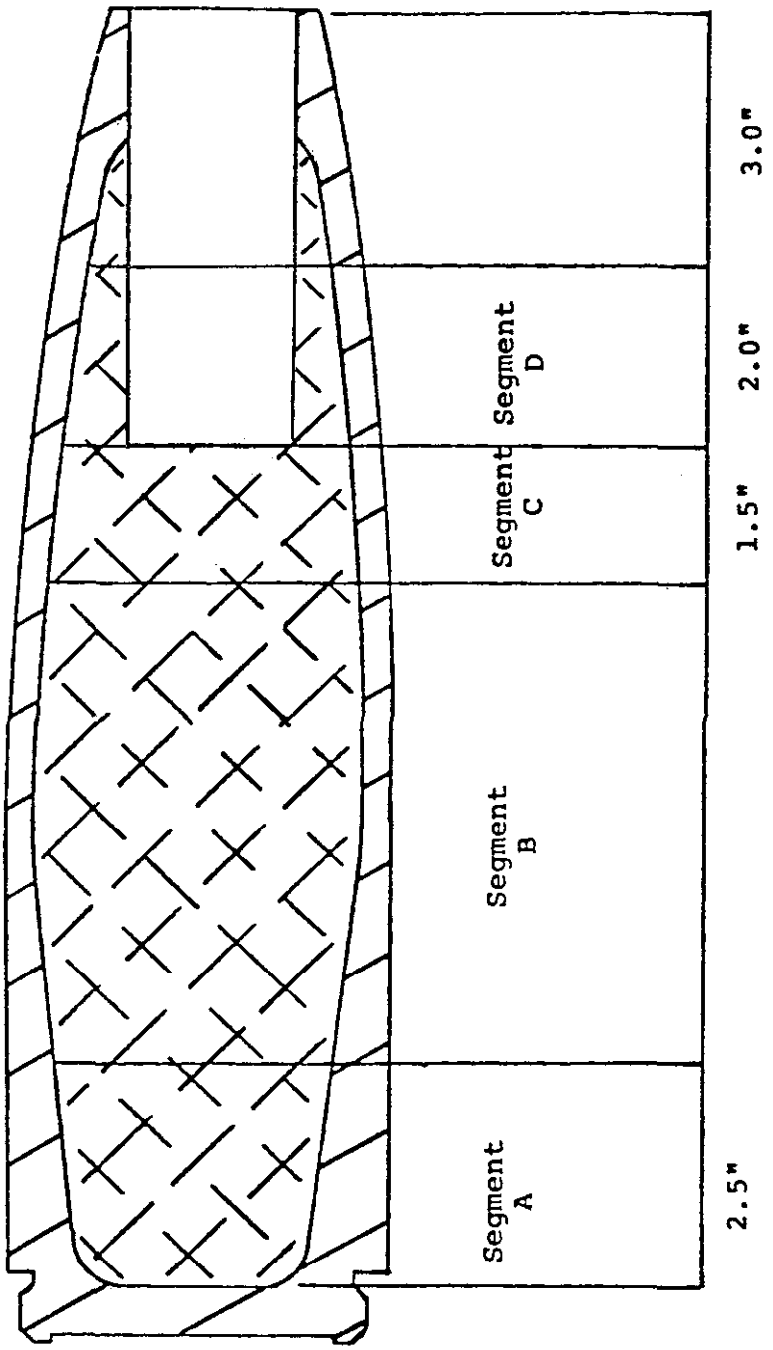


FIGURE 1. Cartridge, 105MM, HERA, M913/M927 Warhead Segments.

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"3.6.2.2 M927 cartridge pressure. The pressure of any individual cartridge, fired at +145 degree F, shall not produce a PIMP greater than 44,800 psi."

PAGE 9

3.10.5 Delete in its entirety.

* Add 3.11 as follows: “3.11 Delay shear seal push-out force. The force required to push a delay out of a rocket motor after pressing of the shear seal shall be 950 pounds to 1300 pounds.”

PAGE 10

4.3.1 Line 11, Delete "Flash reducer, 12630698, 10" in its entirety.

PAGE 11

Table I. Lines 10, 11 and 12, Delete "Flash reducer, (Dwg 126 30698), Examinations for defects, 10, 3.2, 4.4.2.18" in its entirety.

Table I. Change the paragraph reference in note a. from "4.4.2.1" to "4.4.2.2".

PAGE 13

4.4.1.f: Delete "up to four (4) consecutively produced lots of".

4.4.1.h: Delete in its entirety and substitute the following new paragraph:

"h. Delay assemblies from one manufacturer, with constituent chemicals from one lot each, and having the same fuel/oxidizer ratio."

4.4.1.k: Delete “Potassium sulfate from not more than one lot.” in its entirety.

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* 4.4.1.k: Delete j in its entirety and substitute: “j. Shear seals from not more than one lot interfix from one manufacturer.”

4.4.1.l: Delete "M30 propellant from not more than one lot." in its entirety.

PAGES 15 THRU 18

4.4.2.1, 4.4.2.2, 4.4.2.3 and 4.4.2.4; Block, Next Higher Assy Add "/12926808".

PAGES 19 THRU 25

4.4.2.5, 4.4.2.6, 4.4.2.7, 4.4.2.9, 4.4.2.10, and 4.4.2.11; Block, Drawing Number. Add "/12926808".

PAGE 22

4.4.2.8: Delete in its entirety without substitution.

PAGE 23

4.4.2.9: Major 103, Delete "3.2.2" and substitute "3.2.3”.

PAGE 25

4.4.2.11 Add the following Special defect: "1 Weight zone markings, missing or incorrect 100% 3.2 Scale/Visual" Major 107: Change from “Test/4.4.3.6/4.5.2” to “Visual/ (Witness during assembly)/4.4.3.6”.

PAGE 27

4.4.2.13, Block, Drawing Number: Delete "9390990/9391033" and substitute "12938083/8595386 “Block, Next Higher Assembly, add: "9390990/9391033".

PAGE 28

4.4.2.14, Block, Drawing Number: Delete "9205472/9382042" and substitute "12630699/9205472”.

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4.4.2.14 Critical defect #1. Add: "/scale" after "Gage"

PAGE 29

4.4.2.15, Block, Drawing Number: Delete "9391033" and substitute "9390990/9391033".

Under EXAMINATION OR TEST: Change Majors 101 and 102 as follows:
Major 101, Change “Incorrect sequence or increments (M67)” to “Incorrect sequence of increments (M67)”.
Major 102, Change “Foil side of bag, charge 5, not facing primer (M67)” to “Foil side of bag, charge 5, not facing primer (M67), if applicable”.

PAGE 31

4.4.2.17 Majors 101, 102, 103 Add: "/SMTE" after "Gage"

* Add Major Defects 104 and 105 as follows: “104 Evidence of bumps, blisters, pits, splits, delaminations, corrosion, or other defects immediately prior to assembly into motor body 100% 3.1 Visual” “105 Failure to meet push-out force requirments 4.4.3.8 3.11 Test”

PAGE 32, 33 and 34

4.4.2.18, 4.4.2.19, and 4.4.2.20 Delete in their entirety without substitution.

PAGE 35

4.4.2.21 Delete drawing number "12630699" and substitute "12624496".

PAGE 37

4.4.2.23 Delete drawing Number "12934900" Substitute: "12624496"

Add to "Next Higher Assembly" "12934900"

Change title to "Packing and Marking of Sealed Fiber Container and Metal Container Components (prior to sealing of Metal Container)".

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PAGE 38

4.4.2.24 "Next Higher Assembly" Delete Drawing Numbers entirely and substitute: "12929177/12929178"

Add the following Major defects:

"106 Insufficient fillers for tight pack	Level III	3.1	Visual
107 Filler Size (min.)	Level III	3.1	Gage
108 Filler not present underneath cloth strap	Level III	3.1	Visual"

Minor defect 202 delete in its entirety and substitute the following:

"202, Evidence of poor workmanship, Level V 3.10 Visual"

PAGE 41

4.4.3.1.2 Line 6, Delete "6.10" and substitute "6.13".

4.4.3.3 Lines 3 thru 8. Delete second and third sentence"

"Each assembly shall ----------. both channels of the delay assembly. Necessary controls ---------- both channels are visible when radiographed." and substitute the following "Each asssembly shall be subjected to a radiographic examination in a plane that will clearly define all of the Motor Body components."

PAGE 42

4.4.3.4, Add critical defect 3 as follows:

"3 Pressure in excess of 44,800 psi (see 3.6.2.2)"

Major defect 103, delete "Muzzle velocity" and substitute "M913 cartridge muzzle velocity"

4.4.3.4 Change "4.5.3" to "4.5.7".

4.4.3.4 Special a, Delete "6.15" and substitute "6.14".

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PAGE 43

4.4.3.4.1 Change "4.5.8.1" to "4.5.7.1".

4.4.3.4.2 Change "4.5.3.2" to "4.5.7.2".

4.4.3.4.2.3 Line 6 Delete "6.13" and sustitute "6.12".

4.4.3.5 In the second sentence, add a "," after word "change"

PAGE 44

4.4.3.7, Line 12, Delete "6.11" and sustitute "6.16".

* Add 4.4.3.8 as follows: “4.4.3.8 Delay shear seal push-out force qualification test. Prior to assembly of each lot of shear seals into each manufacture’s lot interfix of rocket motor bodies, five sample assemblies shall be pressed with the load, dwell, and press tooling to be used in production. Each sample assembly shall be radio-graphically inspected in accordance with 4.4.3.3. Each sample delay shall be pressed out, and the force required shall be recorded. If any sample falls outside the allowable range, the motor body, seal, load and dwell combination shall not be used in production. Rate of application of test load shall be .5 inches per minute.”

PAGE 48

4.5.2, Line 5, Delete “rocket selector cap and the”.

4.5.4 Change "4.4.3.6" to "4.4.3.5"

4.5.5 Delete in its entirety.

PAGE 49

4.5.6, 4.5.6.1, 4.5.6.1.1 and 4.5.6.1.2 Delete in their entirety.

4.5.7 Change "M557" to "M739 series".

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4.5.7 b. Add the following new subparagraphs:

"6. Chamber pressure (+145 degree firing)
7. Range and deflection of impacts
8. Rocket motor ignition delay time"

PAGE 50

4.5.7 Delete subparagraph c in its entirety without substitution.

4.5.7.1 Delete in its entirety and substitute the following new paragraph:

"4.5.7.1 Functioning/precision/muzzle velocity. Cartridges for this test shall be conditioned at 70 +/- 2 1/2 degrees F for 16 hours (minimum) prior to testing. The M913 test cartridges shall have their muzzle velocity corrected for projectile weight and gun tube wear in accordance with the acceptable M913 firing tables. At least three warmer/spotter cartridges shall be fired prior to firing test cartridges. Time lapses of more than 15 minutes between consecutive firings shall not be permitted without firing additional warmer cartridges. The test shall be fired at 700 to 800 MILS QE. Range probable error shall be calculated for the test cartridges."

"Two master calibration cartridges shall be fired with each 10, M913 test cartridges with muzzle velocity and chamber pressure recorded for information only."

PAGE 51

4.5.7.2 Delete in its entirety and substitute the following new paragraph:

"4.5.7.2 Safety. Cartridges for this test shall be divided into two equal groups and temperature conditioned for 16 hours (minimum) prior to testing. Group 1 shall be conditioned and fired at -50 +5 degrees F. Group 2 shall be conditioned and fired at +145 -5 degrees F. The two groups shall be fired at a convenient elevation selected by the proving ground."

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5.4.1 Change "19048-4231 to "19-48-4231".

PAGE 52

6.3, Line 4, Delete "AMSMC-QAR-I (D)" and substitute "AMSTA-AR-QAA-R".

PAGE 53

6.7, Line 23, Delete "4.3.2" and substitute "4.5.7.2".

6.8, Line 5, Delete "AMSMC-QAR-Q" and substitute "AMSTA-AR-QAA-R".

PAGE 54

6.9.1 Delete in its entirety and substitute the following:

"6.9.1 Radiographic film sources. Approved list of film types and vendors for use in radiographing this projectile:

Eastman Kodak Company
1187 Ridge Road, West
Rochestor, NY 14650

Film Type: AA, T, AX-1, M, MX

Agfa Corporation
NDT Systems
100 Challenger Road
Ridgefield Park, NJ 07660

Film Type: Agfa D7

Fuji Idustrial Radiographic Products
Corporate Headquarters
20 North Ave.
Larchmont, NY 10538

Film Type: Fuji 100"

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6.10 Delete in its entirety and substitute the following new paragraph:

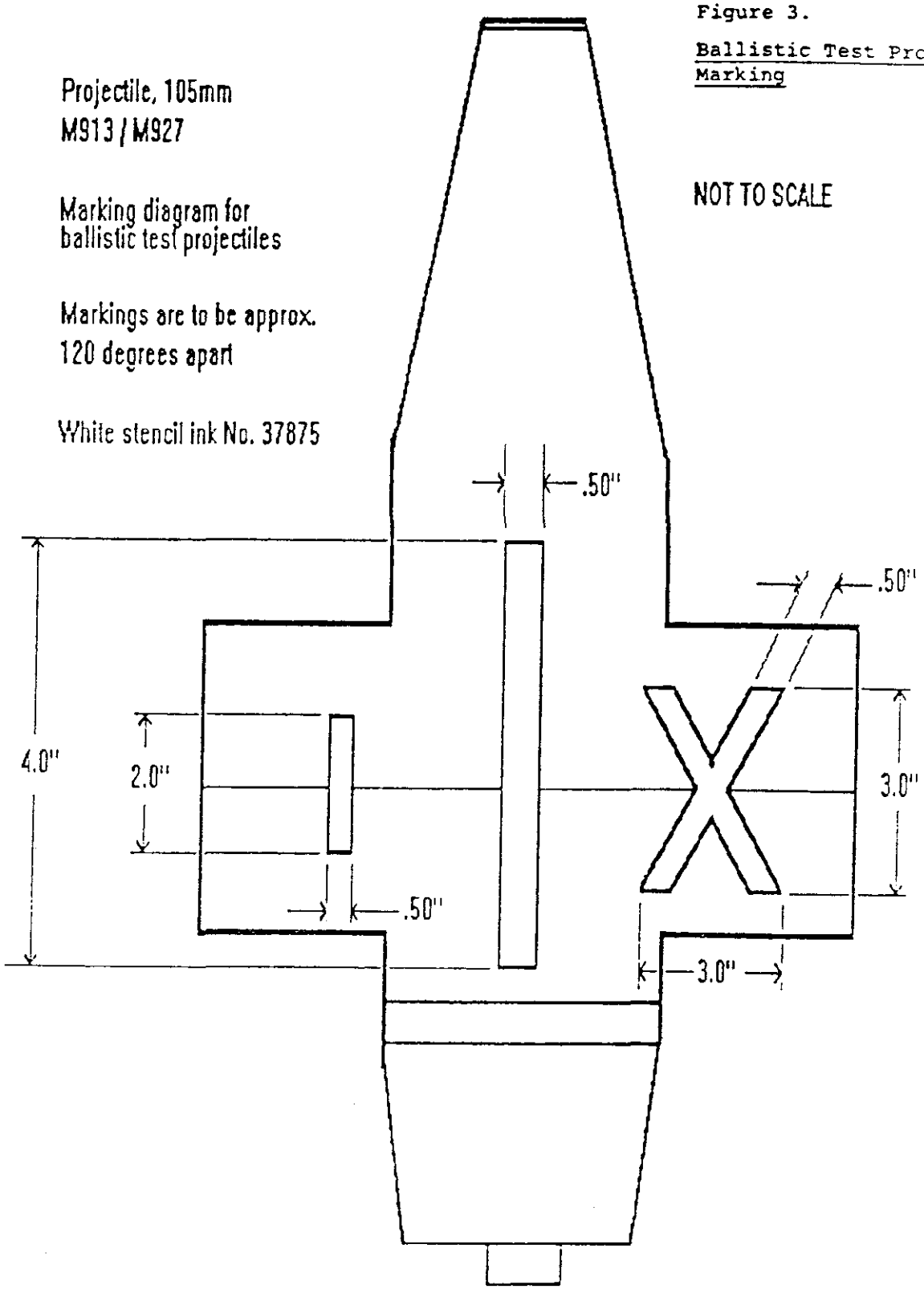
"6.10 Ballistic test projectiles. All projectiles tested by a Government Proving Ground for Ballistic Acceptance Test shall have three markings 120 degrees apart, painted at the LAP manufacturing facility across the warhead/motor body joint. See figure 3."

Add new Figure 3 as included page 54a.

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Figure 3.

Ballistic Test Projectile Marking



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